10_FileMaker_Implementation_Guide.md

MonitorAI v1.5

FileMaker Pro 19 / FileMaker Server 19 Implementation Guide

Version: 1.5 REV2

Status: FINAL BASELINE

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1. Purpose

This document defines how to implement the FileMaker side of MonitorAI v1.5 using:

- FileMaker Pro 19
- FileMaker Server 19
- FileMaker Data API

FileMaker is used as both:

- Clinical monitoring database
- Future AI dataset repository

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2. Environment

Development / Editing:

FileMaker Pro 19

Hosting:

FileMaker Server 19

API:

FileMaker Data API

Storage:

External Secure Container Storage

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3. Required Tables

Create the following tables:

- Patients
- MonitoringSessions
- CaptureImages
- CaptureMetadata
- UploadQueue
- TrainingLabels
- SystemSettings

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4. Primary Key Rule

All primary keys must be UUID text fields.

Recommended primary keys:

- PatientUUID
- SessionUUID
- CaptureUUID
- MetadataUUID
- QueueUUID
- LabelUUID

Auto Enter:

Get (UUID)

Validation:

- Not Empty
- Unique

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5. Patients Table

Purpose:

Patient master reference.

Required Fields:

- PatientUUID
- PatientID
- FirstName
- LastName
- DateCreated
- DateModified

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6. MonitoringSessions Table

Purpose:

One monitoring session per patient per submission.

Required Fields:

- SessionUUID
- PatientUUID
- SessionStartTime
- SessionEndTime
- SessionDurationSeconds
- DeviceModel
- DeviceModelIdentifier
- iOSVersion
- AppVersion
- CalibrationDurationSeconds
- TotalRetakes
- UploadStatus
- UploadStartTime
- UploadCompleteTime
- UploadDurationSeconds
- DateCreated
- DateModified

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7. CaptureImages Table

Purpose:

Store accepted captured images.

One record per accepted image.

Expected:

11 final accepted image records per completed session.

Required Fields:

- CaptureUUID
- SessionUUID
- CaptureStep
- CaptureName
- ImageFile
- ThumbnailFile
- ImageWidth
- ImageHeight
- ImageSizeBytes
- ImageSHA256
- CaptureTimestamp
- CaptureStartTime
- CaptureSuccessTime
- CaptureDurationSeconds
- RetakeVersion
- FinalAccepted

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8. CaptureMetadata Table

Purpose:

Store all metadata related to each image.

Required Fields:

- MetadataUUID
- CaptureUUID
- PTGCurrentDot
- PTGMaxDot
- VisibilityScore
- MaxVisibilityScore
- BuccalVisibilityScore
- OcclusalVisibilityScore
- PlateauTriggered
- PlateauDurationSeconds
- CaptureTriggerType
- CameraPosition
- ActiveCameraDevice
- PreviewZoomLevel

- StoredImageZoomLevel
- TorchEnabled
- TorchLevel
- TorchAdjustmentEvents
- ExposureValue
- BrightnessScore
- WhiteBalanceEstimate
- FocusScore
- BlurScore
- ExposureScore
- CompositionScore
- CalibrationOffsetX
- CalibrationOffsetY
- CalibrationQualityScore
- VoiceEventCount
- VoiceEvents
- HapticEventCount
- HapticEvents
- UploadStartTime
- UploadCompleteTime
- UploadDurationSeconds
- AppBuildNumber
- BatteryLevel
- NetworkType
- UploadAttemptCount

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9. UploadQueue Table

Purpose:

Offline upload recovery.

Required Fields:

- QueueUUID
- CaptureUUID
- QueuedTime
- RetryCount
- LastRetryTime
- NextRetryTime
- QueueStatus
- LastErrorCode
- LastErrorMessage

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10. TrainingLabels Table

Purpose:

Future AI annotation and dataset training.

Required Fields:

- LabelUUID
- CaptureUUID
- UseForTraining
- AlignerFit
- AnnotationStatus
- DoctorComment
- AnnotatedBy
- AnnotatedAt

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11. SystemSettings Table

Purpose:

Global MonitorAI configuration.

Required Fields:

- TorchDefaultLevel
- OverlayOpacity
- AutoCaptureSeconds
- PlateauCaptureSeconds
- MinimumPTGForCapture
- PreviewZoomNormal
- PreviewZoomLarge
- PreviewZoomExtraLarge

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12. Relationships

Create the following relationships:

Patients::PatientUUID

=

MonitoringSessions::PatientUUID

MonitoringSessions::SessionUUID

=

CaptureImages::SessionUUID

CaptureImages::CaptureUUID

=

CaptureMetadata::CaptureUUID

CaptureImages::CaptureUUID

=

UploadQueue::CaptureUUID

CaptureImages::CaptureUUID

=

TrainingLabels::CaptureUUID

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13. Container Field Configuration

Container fields:

- CaptureImages::ImageFile
- CaptureImages::ThumbnailFile

Storage:

External Secure Storage

Recommended:

Do not embed images inside the FileMaker file.

Reason:

- Smaller FileMaker file size
- Better backup performance
- Better dataset export capability
- Better future AI workflow compatibility

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14. Image Integrity Fields

For every accepted image, store:

- ImageWidth
- ImageHeight
- ImageSizeBytes
- ImageSHA256

Purpose:

- Dataset validation
- Duplicate detection
- Export integrity
- Future AI training traceability

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15. API Layouts

Create dedicated layouts only for Data API access.

Do not use clinical layouts as API layouts.

Required API Layouts:

- MonitorAI_API_Sessions
- MonitorAI_API_Captures
- MonitorAI_API_Metadata
- MonitorAI_API_UploadQueue
- MonitorAI_API_TrainingLabels
- MonitorAI_API_SystemSettings

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16. API Layout Rules

API layouts should contain only fields required for API communication.

Do not include:

- Portals
- Buttons
- Web Viewers
- Unnecessary calculations
- Script triggers

Purpose:

- Improve Data API performance
- Reduce unexpected behavior
- Simplify debugging

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17. FileMaker Data API Configuration

In FileMaker Server 19 Admin Console:

Enable FileMaker Data API.

Confirm HTTPS access.

Confirm SSL certificate is valid.

Confirm database is hosted and accessible.

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18. API Account

Create a dedicated account.

Recommended Account Name:

monitorai_api

Do not use an admin account.

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19. Privilege Set

Create a dedicated privilege set.

Recommended Name:

MonitorAI_API_Privileges

Allow:

- View records
- Create records
- Edit records
- Container upload
- Data API access

Deny:

- Delete records
- Layout editing
- Script editing
- Account management
- Full access

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20. Value Lists

Create required value lists:

- UploadStatus
- CaptureStep
- CaptureName
- PTGDot
- CaptureTriggerType
- CameraPosition
- PreviewZoomLevel
- QueueStatus
- AlignerFit
- AnnotationStatus
- NetworkType

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21. Recommended Indexes

Index the following fields:

Patients::PatientUUID

MonitoringSessions::SessionUUID

MonitoringSessions::PatientUUID

CaptureImages::CaptureUUID

CaptureImages::SessionUUID

CaptureImages::CaptureStep

CaptureImages::CaptureName

CaptureImages::ImageSHA256

CaptureMetadata::MetadataUUID

CaptureMetadata::CaptureUUID

CaptureMetadata::PTGMaxDot

CaptureMetadata::CaptureTriggerType

UploadQueue::QueueStatus

TrainingLabels::UseForTraining

TrainingLabels::AnnotationStatus

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22. Recommended Scripts

Create optional helper scripts:

- API_CreateSession
- API_CreateCapture
- API_CreateMetadata
- API_UpdateSessionStatus
- API_CreateTrainingLabel

- System_GetSettings
- Admin_CheckUploadQueue
- Admin_ExportTrainingDataset

Scripts are optional for basic Data API operation but useful for maintenance and admin workflows.

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23. Data API Workflow

Recommended upload order:

1. Create Session record
2. Create Capture record
3. Create Metadata record
4. Upload ImageFile container
5. Upload ThumbnailFile container
6. Repeat for 11 captures
7. Update Session as Uploaded

Reason:

Metadata is created before image upload so capture context is preserved even if container upload fails.

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24. Backup Strategy

FileMaker Server 19 backup policy:

Recommended:

- Hourly progressive backup
- Daily full backup
- Weekly archive backup
- Minimum 90-day retention

External secure container storage must be included in backup strategy.

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25. Performance Rules

Avoid on API layouts:

- Heavy summary fields
- Unstored calculations

- Portals
- Web Viewers
- Script triggers

Optimize for:

- Fast record creation
- Fast container upload
- Reliable metadata upload

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26. Security Rules

Required:

- HTTPS
- Valid SSL certificate
- Dedicated API account
- External secure container storage
- Least-privilege account design

Prohibited:

- Guest access
- Shared admin accounts
- Plain text credentials
- Patient data in logs

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27. AI Dataset Rules

Never overwrite historical image or metadata records.

Maintain:

- Original image
- Retake history
- Metadata history
- Upload history
- Annotation history

Future AI training requires complete traceability.

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28. Migration Rules

Future schema changes should prefer:

Add fields

rather than

Rename or delete fields.

Reason:

Preserve API compatibility.

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29. Final Principle

FileMaker Server 19 is not only storage.

It is the central repository for:

- Clinical monitoring
- Image metadata
- Upload traceability
- Future AI training datasets

The FileMaker implementation must preserve:

- Data integrity
- Metadata integrity
- Historical traceability
- AI dataset value